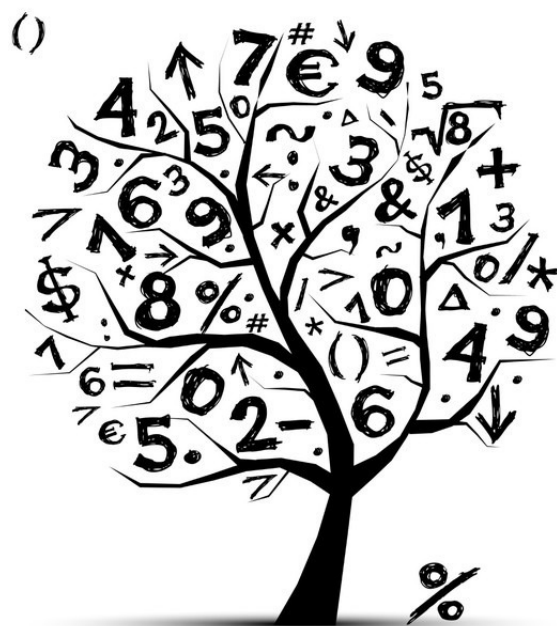


Last update: 7 ottobre 2026

Factorization of matrices



Indice

1 LU factorization

2

1 LU factorization

- The LU factorization (or LU decomposition) is a fundamental matrix decomposition that expresses a matrix as the product of a lower triangular matrix and an upper triangular matrix.

Given a square matrix $A \in \mathbb{R}^{n \times n}$, a **PLU factorization** of A is a decomposition of the form

$$PA = LU,$$

where:

- $P \in \mathbb{R}^{n \times n}$ is a **permutation matrix** (it accounts for possible row swaps),
- $L \in \mathbb{R}^{n \times n}$ is a **lower triangular matrix** with ones on the diagonal,
- $U \in \mathbb{R}^{n \times n}$ is an **upper triangular matrix**.

- The factorization is computed by Gaussian elimination.
- If no row swaps are needed, then $P = I_n$ and the factorization reduces to the **LU factorization**:

$$A = LU.$$

- The PLU/LU factorization is very useful for solving linear systems $Ax = b$ efficiently.

Method to compute the PLU (LU with pivoting) factorization.

Starting from A , we perform Gaussian elimination. Whenever a row swap is required, it is recorded in the permutation matrix P . At the end of the elimination we obtain the upper triangular matrix U , while L stores the elimination multipliers.

Specifically:

- Each elimination step $R_i \leftarrow R_i - \ell_{ij}R_j$ (with $i > j$) creates a zero in position (i, j) .
- The multiplier ℓ_{ij} is stored in position (i, j) of L .
- The diagonal of L is filled with ones.

Esempio 1: LU factorization

Let us compute the LU factorization of

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix}.$$

Step 1: Eliminate the entries below the first pivot $a_{11} = 2$.

We need to eliminate $a_{21} = 4$. The multiplier is:

$$\ell_{21} = \frac{a_{21}}{a_{11}} = \frac{4}{2} = 2.$$

We perform $R_2 \leftarrow R_2 - 2R_1$:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ -2 & 7 & 2 \end{pmatrix}.$$

We need to eliminate $a_{31} = -2$. The multiplier is:

$$\ell_{31} = \frac{a_{31}}{a_{11}} = \frac{-2}{2} = -1.$$

We perform $R_3 \leftarrow R_3 - (-1)R_1 = R_3 + R_1$:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{pmatrix}.$$

Step 2: Eliminate the entries below the second pivot -8 .

We need to eliminate the entry in position $(3, 2)$, which is currently 8. The multiplier is:

$$\ell_{32} = \frac{8}{-8} = -1.$$

We perform $R_3 \leftarrow R_3 - (-1)R_2 = R_3 + R_2$:

$$U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Esempio 2: LU factorization

Step 3: Build the matrix L .

The matrix L has ones on the diagonal and the multipliers below the diagonal:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}.$$

Verification: We check that $LU = A$:

$$\begin{aligned} LU &= \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2-8 & 2-2 \\ -2 & -1+8 & -1+2+1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix} = A. \end{aligned}$$

Esempio 3: PLU factorization with row permutation

Consider the matrix

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Since $a_{11} = 0$, we cannot use it as a pivot. We need to swap rows 1 and 2.

Let

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

be the permutation matrix that swaps rows 1 and 2.

Then:

$$PA = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Now PA already has an upper triangular form (no elimination needed below the diagonal), so:

$$L = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I, \quad U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Therefore, the PLU factorization is:

$$PA = LU,$$

where $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $L = I$, and $U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.